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Integrated circuit and semiconductor module

Abstract

An integrated circuit includes: a power supply line configured to receive a power supply voltage; a constant current source electrically coupled to the power supply line; a reference voltage circuit electrically coupled to the constant current source; and a first resistor having two ends, one end thereof being electrically coupled to the constant current source, and the other end thereof being electrically coupled to the reference voltage circuit. The reference voltage circuit is a bandgap circuit including a plurality of bipolar devices. The first resistor is configured to decrease a leakage current in the bipolar devices when a temperature thereof rises.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This is a continuation application of International Patent Application No. PCT/JP2022/016144 filed Mar. 30, 2022, which claims the benefit of priority to Japanese Patent Application No. 2021-082486 filed May 14, 2021, the entire contents of each of which the entire contents of each of which are incorporated herein by reference.

BACKGROUND

Technical Field

(1) The present disclosure relates to an integrated circuit and a semiconductor module.

Description of the Related Art

(2) Reference voltage circuits utilizing bandgap voltage of semiconductor are known (see, for example, Japanese Patent Application Publication No. 2005-202704).

(3) In bandgap reference voltage circuits including bipolar transistors, when the temperature of such a bipolar transistor rises, leakage current may be generated between an N-well region covering a collector terminal and a substrate. Accordingly, the current outputted from the substrate of such a reference voltage circuit may increase, and the voltage outputted from the reference voltage circuit may also rise.

SUMMARY

(4) A first aspect of an embodiment of the present disclosure is to provide an integrated circuit comprising: a power supply line configured to receive a power supply voltage; a constant current source electrically coupled to the power supply line; a reference voltage circuit electrically coupled to the constant current source; and a first resistor having two ends, one end thereof being electrically coupled to the constant current source, and the other end thereof being electrically coupled to the reference voltage circuit, wherein the reference voltage circuit is a bandgap circuit including a plurality of bipolar devices, and the first resistor is configured to decrease a leakage current in the plurality of bipolar devices when a temperature rises.

(5) A second aspect of an embodiment of the present disclosure is to provide a semiconductor module comprising the integrated circuit.

(6) Note that the above summary does not list all of the features of the present disclosure. Subcombinations of these feature groups may also be the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an example of a configuration of a semiconductor module **10**.
- (2) FIG. 2 illustrates an example of a circuit diagram of a voltage generator circuit **40a**.
- (3) FIG. 3 illustrates an example of a circuit diagram of a reference voltage circuit **53**.
- (4) FIG. 4 is an example of a conceptual diagram illustrating generation of leakage current in a bipolar transistor **87** of a reference voltage circuit **53**.
- (5) FIG. 5 is an example of a schematic diagram illustrating change with temperature in currents and voltages in a voltage generator circuit **40a**.
- (6) FIG. 6 illustrates an example of a circuit diagram of a voltage generator circuit **110** according to a comparative example.
- (7) FIG. 7 is an example of a schematic diagram illustrating change with temperature in currents and voltages in a voltage generator circuit **110**.
- (8) FIG. 8 is an example of a schematic diagram illustrating change with temperatures of current I_m of a voltage generator circuit **40a** and current I_{bg} of a voltage generator circuit **110**.
- (9) FIG. 9A illustrates an example of a circuit diagram of a voltage generator circuit **40c**.
- (10) FIG. 9B illustrates an example of a circuit diagram of a voltage generator circuit **40d**.
- (11) FIG. 10A illustrates an example of a circuit diagram of a voltage generator circuit **40e**.
- (12) FIG. 10B illustrates an example of a circuit diagram of a voltage generator circuit **40f**.
- (13) FIG. 10C illustrates an example of a circuit diagram of a voltage generator circuit **40g**.

DETAILED DESCRIPTION

(14) Hereinafter, the present disclosure will be described through embodiments of the disclosure. However, the following embodiments do not limit the disclosure according to the scope of the claims. In addition, not all the combinations of the features described in the embodiments are necessarily essential for solution in the disclosure.

(15) Herein, the terms “electrically couple” and “couple” are used, and the term “couple” means to “electrically couple” unless otherwise noted.

EMBODIMENT EXAMPLES

Configuration Example of Semiconductor Module **10**

- (16) FIG. 1 illustrates an example of a configuration of a semiconductor module **10**. The semiconductor module **10** is a module to drive a load **11** in response to an instruction from an externally provided microcomputer (not illustrated).
- (17) The semiconductor module **10** uses an externally provided power supply **12** as a main power supply. The power supply **12** applies a voltage HV to the semiconductor module **10**. The semiconductor module **10** includes semiconductor chips **20a**, **20b**, power supplies **21a**, **21b**, and the integrated circuits **22a**, **22b**.
- (18) Here, among the chips and the like configuring the semiconductor module **10**, the semiconductor chip **20a**, the power supply **21a**, and the integrated circuit **22a** are provided on the low side, and the semiconductor chip **20b**, the power supply **21b**, and the integrated circuit **22b** are provided on the high side. Further, in an embodiment of the present disclosure, the circuits on the low side and the circuits on the high side are the same in configuration, and thus the following mainly describes the circuits on the low side.
- (19) The load **11** is, for example, a motor coil, and is driven by a voltage V_{out} outputted from a node of contact between the semiconductor chips **20a** and **20b**.
- (20) The semiconductor chip **20a** includes a switching device to drive the load **11** and a

temperature sensing element. The semiconductor chip **20a** according to an embodiment of the present disclosure includes an insulated gate bipolar transistor (IGBT) **30a** as the switching device, and a diode **31a** for the IGBT **30a** as the temperature sensing element.

(21) However, the switching device provided in the semiconductor chip **20a** is not limited to the IGBT **30a**, and the switching device may be a metal-oxide-semiconductor (MOS) transistor or a bipolar transistor. Further, the semiconductor chip **20a** may include a diode to communicate load current to the IGBT **30a**.

(22) The power supply **21a** is a power supply for the integrated circuit **22a**, and applies the power supply voltage V_{cc1} to a power supply line **L1a**. Note that the power supply **21a** according to an embodiment of the present disclosure is generated by a power supply circuit (not illustrated) provided inside the semiconductor module **10**, but may be supplied from the outside, for example.

(23) The integrated circuit **22a** is a low-voltage integrated circuit (LVIC), and outputs a driving signal **LO** to the gate electrode of the IGBT **30a** in response to signal **LIN** inputted from a microcomputer (not illustrated), to thereby control the IGBT **30a**. The integrated circuit **22a** includes a voltage generator circuit **40a**, a temperature detection circuit **41a**, and a control circuit **42a**.

(24) The voltage generator circuit **40a** according to an embodiment of the present disclosure generates a reference voltage V_{ref1} , based on the power supply voltage V_{cc1} of the power supply line **L1a**.

(25) The temperature detection circuit **41a** supplies predetermined current to the diode **31a**, and outputs a temperature sensing signal T_{sns1} corresponding to the temperature of the IGBT **30a** to the control circuit **42a**, based on the forward voltage of the diode **31a**.

(26) The control circuit **42a** controls the operation of the IGBT **30a**, based on the signal **LIN** from the microcomputer (not illustrated), the reference voltage V_{ref1} , and the temperature sensing signal T_{sns1} .

(27) Specifically, the control circuit **42a** controls switching of the IGBT **30a** in response to the signal **LIN**. In addition, the control circuit **42a** detects overheating of the semiconductor chip **20a**, based on the reference voltage V_{ref1} and the temperature sensing signal T_{sns1} . When detecting overheating of the semiconductor chip **20a**, the control circuit **42a** protects the IGBT **30a** from heat by turning off the IGBT **30a**, for example.

(28) The semiconductor chip **20b** includes an IGBT **30b** and a diode **31b**, as corresponding components on the high side, and the power supply **21b** applies a power supply voltage V_{cc2} to a power supply line **L1b**. Further, the integrated circuit **22b** includes a voltage generator circuit **40b**, a temperature detection circuit **41b**, and a control circuit **42b**.

(29) The voltage generator circuit **40b** supplies a reference voltage V_{ref2} to the control circuit **42b**, as in the voltage generator circuit **40a**. The temperature detection circuit **41b** outputs a temperature sensing signal T_{sns2} corresponding to the temperature of the IGBT **30b** to the control circuit **42b**, based on the forward voltage of the diode **31b**, as in the temperature detection circuit **41a**. The control circuit **42b** controls the operation of the IGBT **30b**, based on a signal **HIN** from the microcomputer (not illustrated), the reference voltage V_{ref2} , and the temperature sensing signal T_{sns2} . The control circuit **42b** includes a level converter circuit to convert the signal **HIN** whose reference voltage is **GND** into a signal whose reference voltage is V_{out} .

(30) As such, the voltage generator circuit **40b** and the temperature detection circuit **41b** have the same functions and configurations as the voltage generator circuit **40a** and the temperature detection circuit **41a** have, respectively. Accordingly, the description of the integrated circuit **22b**, which includes the voltage generator circuit **40b**, the temperature detection circuit **41b**, and the control circuit **42b** on the high side, is omitted below.

Configuration Example of Voltage Generator Circuit **40a**

(31) FIG. 2 illustrates an example of a circuit diagram of the voltage generator circuit **40a**. The voltage generator circuit **40a** generates the temperature-compensated reference voltage V_{ref1} of a

predetermined level. The voltage generator circuit **40a** according to an embodiment of the present disclosure includes a bias current source **50a**, a current mirror circuit **51**, a resistor **52**, and a reference voltage circuit **53**.

(32) The bias current source **50a** generates a predetermined bias current I_{bias} . A bias current source **50** according to an embodiment of the present disclosure includes a depletion type MOS transistor **61** whose gate terminal and source terminal are diode-coupled.

(33) Here, the bias current source **50a** is configured with a single device, in other words, the depletion type MOS transistor **61**. Accordingly, with the use of the bias current source **50a** according to an embodiment of the present disclosure, the circuit size of the voltage generator circuit **40a** is reduced.

(34) The current mirror circuit **51** supplies constant current I_m to the reference voltage circuit **53**, based on the bias current I_{bias} . The current mirror circuit **51** is electrically coupled to the power supply line **L1a**, to which the power supply voltage V_{cc1} is applied. The current mirror circuit **51** according to an embodiment of the present disclosure includes a MOS transistor **70**, through which the bias current I_{bias} flows, and a MOS transistor **71**. Note that the MOS transistors **70**, **71** according to an embodiment of the present disclosure are P-channel metal-oxide-semiconductor (PMOS) transistors.

(35) The gate terminal and source terminal of the MOS transistor **70** are diode-coupled. The gate terminal of the MOS transistor **70** and the gate terminal of the MOS transistor **71** are electrically coupled each other. Accordingly, based on the bias current I_{bias} flowing through the MOS transistor **70**, the current I_m is supplied from the MOS transistor **71**.

(36) The resistor **52** generates voltage V_{dd1} to operate the reference voltage circuit **53**, from the current I_m of the current mirror circuit **51**. The resistor **52** has one end electrically coupled to the current mirror circuit **51**, and the other end electrically coupled to the reference voltage circuit **53**.

(37) The reference voltage circuit **53** outputs the reference voltage V_{ref1} , which is to be used in other circuits, based on the inputted current I_m and voltage V_{dd1} . Note that, although details will be described later, in the reference voltage circuit **53** according to an embodiment of the present disclosure, a node different from a node from which the reference voltage V_{ref1} is to be outputted is grounded. Accordingly, when leakage current is not generated in the reference voltage circuit **53**, the current I_{bg} results in the current I_m .

(38) Note that the current mirror circuit **51** according to an embodiment of the present disclosure corresponds to a “constant current source”, and the current I_m corresponds to “first current”. Further, the MOS transistor **70** corresponds to a “first MOS transistor”, the MOS transistor **71** corresponds to a “second MOS transistor”. Further, the resistor **52** corresponds to a “first resistor”.

(39) Here, a description has been given, as an example, of the voltage generator circuit **40a** on the low side, which uses the power supply voltage V_{cc1} , the power supply line **L1a**, GND as a power supply reference voltage, the power supply line **L1a** and the reference voltage V_{ref1} to be outputted. The voltage generator circuit **40b** on the high side also has a similar configuration using the power supply voltage V_{cc2} , the power supply line **L1b**, V_{out} as a power supply reference voltage, and the reference voltage V_{ref2} to be outputted in place of those on the low side.

Configuration Example of Reference Voltage Circuit **53**

(40) FIG. 3 illustrates an example of a circuit diagram of the reference voltage circuit **53**. The reference voltage circuit **53** according to an embodiment of the present disclosure includes MOS transistors **81** to **84**, a resistor **85**, and bipolar transistors **86**, **87**. In other words, the reference voltage circuit **53** according to an embodiment of the present disclosure is a bandgap circuit including bipolar devices. Note that the MOS transistors **81**, **82** according to an embodiment of the present disclosure are PMOS transistors, and the MOS transistors **83**, **84** are NMOS transistors.

(41) The gate terminal and source terminal of each of the MOS transistors **82**, **83** are diode coupled. The MOS transistors **81**, **82** configure a P-channel current mirror circuit, and the MOS transistors **83**, **84** configure an N-channel current mirror circuit.

(42) In response to the current I_m being supplied to the diode-coupled MOS transistor **82** from the current mirror circuit **51**, the MOS transistor **81** is turned on. Accordingly, the MOS transistor **81** outputs current, based on the current flowing through the MOS transistor **82**. As a result, the MOS transistors **81**, **82** supply current to the MOS transistors **83**, **84**, respectively.

(43) Further, in response to the current being supplied to the diode-coupled MOS transistor **83**, the MOS transistor **84** is turned on. Accordingly, the MOS transistor **84** outputs current, based on the current flowing through the MOS transistor **83**. As a result, the MOS transistor **83** supplies current to the bipolar transistor **86**, and the MOS transistor **84** supplies current to the resistor **85**.

(44) Further, in an embodiment of the present disclosure, the MOS transistors **81**, **82** have the same size, and the MOS transistors **83**, **84** have the same size. Accordingly, the currents from the current mirror circuit including the MOS transistors **83**, **84** are equal to each other.

(45) Accordingly, the current I_m from the current mirror circuit **51** is supplied to the MOS transistors **81**, **83**, and the MOS transistors **81**, **83** can be considered as configuring a current source **88** to supply current to the bipolar transistor **86**. The current source **88** according to an embodiment of the present disclosure corresponds to a “first current source”.

(46) Similarly, the current I_m from the current mirror circuit **51** is supplied to the MOS transistors **82**, **84**, and the MOS transistors **82**, **84** can be considered as configuring a current source **89** to supply current to the resistor **85**. Further, the current source **89** is electrically coupled in parallel with the current source **88**. The current source **89** according to an embodiment of the present disclosure corresponds to a “second current source”.

(47) Here, the current source **88**, **89** is a current source in which the magnitude of the current to be outputted changes with the level of the power supply voltage V_{dd1} applied to the current source **88**, **89**, unlike such a “constant current source” as the current mirror circuit **51** configured to generate the current I_m , based on the bias current I_{bias} . In other words, herein, the “current source” is different from the “constant current source” in that although current is supplied, the magnitude of the current to be supplied is not constant.

(48) The resistor **85** has one end coupled to the output of the MOS transistor **84**, and the other end coupled to the collector terminal of the bipolar transistor **87**. The current from the MOS transistor **84** is supplied to the resistor **85**, and the resistor **85** is electrically coupled to the bipolar transistor **87**.

(49) The base terminal and collector terminal of the bipolar transistor **86** are electrically coupled and the base terminal and collector terminal of the bipolar transistor **87** are electrically coupled, and the emitter terminals of the bipolar transistors **86**, **87** are grounded.

(50) The bipolar transistors **86**, **87** are respectively configured such that the base-emitter voltages thereof are different from each other. Specifically, the bipolar transistor **86** according to an embodiment of the present disclosure includes a single bipolar transistor, meanwhile the bipolar transistor **87** includes a plurality of bipolar transistors coupled in parallel. Accordingly, the base-emitter voltage of the bipolar transistor **86** is greater than the base-emitter voltage of the bipolar transistor **87**. Note that both of the base-emitter voltages of the bipolar transistors **86**, **87** have positive temperature coefficients.

(51) Further, in an embodiment of the present disclosure, the currents from the current sources **88**, **89** are equal to each other, which results in the voltages at source terminals of the MOS transistors **83**, **84** being equal to each other. Accordingly, voltage with a negative temperature coefficient is generated across the resistor **85**, in accordance with the difference between the base-emitter voltage of the bipolar transistor **86** and the base-emitter voltage of the bipolar transistor **87**.

(52) As a result, the voltage obtained by adding the base-emitter voltage of the bipolar transistor **87** with a positive temperature coefficient and the voltage across the resistor **85** with a negative temperature coefficient is generated, as the reference voltage V_{ref1} , at the node at which the MOS transistor **84** and the resistor **85** are coupled. Note that, in an embodiment of the present disclosure, for example, the resistance value of the resistor **85** and the number of the bipolar transistors **87** are

adjusted such that the temperature coefficient of the reference voltage V_{ref1} is zero.

(53) Accordingly, the temperature-compensated reference voltage V_{ref1} is supplied from the reference voltage circuit **53**.

(54) The resistor **85** according to an embodiment of the present disclosure corresponds to a “second resistor”. Further, the bipolar transistor **86** corresponds to a “first bipolar transistor”, and the bipolar transistor **87** corresponds to a “second bipolar transistor”.

(55) As described above, the reference voltage circuit **53** according to an embodiment of the present disclosure is capable of outputting the temperature-compensated reference voltage V_{ref1} . Incidentally, when a temperature rises, the leakage current is generated from the bipolar transistors **86**, **87**, which may cause the reference voltage V_{ref1} to fluctuate greatly in accordance with the temperature. The following describes the leakage current generated in the bipolar transistors **86**, **87**. Leakage Current Generated in Bipolar Transistor

(56) FIG. **4** is an example of a conceptual diagram illustrating generation of the leakage current in the bipolar transistor **87** of the reference voltage circuit **53**. In an embodiment of the present disclosure, the bipolar transistor **87** is formed such that an N-well region **92** is provided in a substrate **91** in the semiconductor, and a dopant diffusion region for each terminal to function is provided in the N-well region **92**. In an embodiment of the present disclosure, the substrate **91** has P-type conductivity.

(57) An N+ collector region **93** is provided around a region in which a collector terminal **96** is provided in the N-well region **92**, and a P+ base region **94** is provided around a region in which a base terminal **97** is provided. Further, an N+ emitter region **95** is provided around a region in which an emitter terminal **98** is provided in the P+ base region **94**.

(58) Note that, in a region marked with each conductivity type, a region prefixed with “+” means that a doping concentration therein is higher than that in a region without “+”, and a region prefixed with “-” means that a doping concentration therein is lower than that in a region without

(59) In the bipolar transistor **87** as such, the surface area of a PN junction portion between the substrate **91** and the N-well region **92** is large. When the temperature of a semiconductor device rises, the larger the surface area of the PN junction portion, the greater the possibility of the leakage current being generated. Accordingly, in the bandgap reference voltage circuit **53** using bipolar devices, it is effective to reduce the leakage current.

(60) In the bipolar transistor **87**, the current flowing from the collector terminal **96** into the bipolar transistor is reduced, thereby being able to restrain the leakage current even when the temperature rises. Accordingly, the current I_m inputted to the reference voltage circuit **53** is reduced and the voltage V_{dd1} is lowered, thereby restraining the leakage current.

(61) Note that the bias current source **50a** and the current mirror circuit **51** according to an embodiment of the present disclosure include no bipolar device, and thus the voltage generator circuit **40a** is configured such that the leakage current is less likely to be generated from those other than the reference voltage circuit **53**.

(62) Further, the bipolar transistor **87** is used as an example, to explain the mechanism of the leakage current in the bipolar device in the reference voltage circuit **53**, however, the leakage current may be generated based on a similar mechanism, in the bipolar transistor **86** as well. As such, in an embodiment of the present disclosure, the term “the leakage current” indicates the current flowing from the N-well region **92** to the substrate **91** when a bipolar transistor is formed, for example.

Change with Temperature in Current and Voltage Values In Voltage Generator Circuit **40a** of Embodiment Example

(63) FIG. **5** is an example of a schematic diagram illustrating change with temperature in current values and voltage values in the voltage generator circuit **40a**. An embodiment of the present disclosure illustrates the relationship among the power supply voltage V_{cc1} , current I_{bg} , the current I_m , and the reference voltage V_{ref1} when a temperature T [$^{\circ}$ C.] of the reference voltage

circuit **53** is changed.

(64) The current I_{bg} is the current flowing from the reference voltage circuit **53** to the ground. The substrate **91** of the bipolar transistor **87** is grounded, and thus when the leakage current flowing from the substrate **91** to the ground explained with reference to FIG. **4** increases, the current I_{bg} increases.

(65) Note that the substrate of the bipolar transistor **86** is also grounded, and thus even when the leakage current in the bipolar transistor **86** increases as well, the current I_{bg} increases.

(66) A temperature threshold value T_{th} is a temperature value at which the leakage current flowing from the bipolar device of the reference voltage circuit **53** to the ground is generated. Here, in an embodiment of the present disclosure, the phrase “the leakage current is generated” indicates, for example, that the value of the current flowing from the N-well region **92** when the bipolar transistor is formed to the substrate **91** (hereinafter, referred to as current I_x) reaches a predetermined multiple (e.g., 5 times) of the current I_x when a temperature is a predetermined temperature (e.g., 25° C.)

(67) In an embodiment of the present disclosure, the temperature threshold value T_{th} at which the leakage current is generated is 100° C., however, it varies depending on the configuration of the bipolar transistor and/or the dopant concentration in the substrate **91** and/or the N-well region **92**. Accordingly, the temperature threshold value T_{th} may not be 100° C., but may be other temperatures such as 120° C. or the like.

(68) In the voltage generator circuit **40a**, the small constant current I_m based on the bias current I_{bias} is provided from the current mirror circuit **51** to the reference voltage circuit **53**. Accordingly, even in a temperature range above the temperature threshold value T_{th} , the leakage current from the reference voltage circuit **53** is restrained, and the current I_{bg} results in being substantially the same as the current I_m . The current I_m is determined such that the current I_{bg} is limited to the value of the current I_m , when the temperature becomes equal to or higher than the temperature threshold value T_{th} at which the leakage current flowing from the reference voltage circuit **53** to the ground is generated.

(69) Even when the temperature exceeds the temperature threshold value T_{th} , the value of the current I_{bg} is substantially the same as the value of the current I_m . As a result, the current flowing through the resistor **85** is substantially constant, irrespective of temperature. Accordingly, even when the reference voltage circuit **53** changes in temperature, the reference voltage V_{ref1} generated from the reference voltage circuit **53** is also substantially constant.

(70) Note that, in an embodiment of the present disclosure, the resistor **52** is provided between the current mirror circuit **51** and the reference voltage circuit **53**. Accordingly, the voltage V_{dd1} supplied to the reference voltage circuit **53** is lower than that when the resistor **52** is not provided. This lowers the voltages at the nodes in the reference voltage circuit **53**, and lowers the voltages applied to the collector terminals of the bipolar transistors **86**, **87** as well.

(71) The lower the voltages applied to the collector terminals of the bipolar transistors **86**, **87**, the smaller the leakage current. Accordingly, in an embodiment of the present disclosure, by lowering the voltage V_{dd1} applied as power supply for the reference voltage circuit **53**, the leakage current can be reduced smaller. This makes it possible to generate the temperature-compensated reference voltage V_{ref1} with high accuracy.

Comparative Example

Voltage Generator Circuit **110** of Comparative Example

(72) FIG. **6** illustrates an example of a circuit diagram of a voltage generator circuit **110** according to a comparative example. The following mainly describes differences between the voltage generator circuit **110** and the voltage generator circuit **40a**.

(73) The voltage generator circuit **110** of a comparative example does not include the current mirror circuit **51** or the resistor **52**. In other words, the reference voltage circuit **53** of the voltage generator circuit **110** according to a comparative example is directly electrically coupled to the

power supply line **L1a** to supply the power supply voltage **Vcc1**.

(74) Current **Icc1** is supplied from the power supply line **L1a** to the reference voltage circuit **53**. The current **Icc1** corresponds to a “second current”.

(75) FIG. 7 is an example of a schematic diagram illustrating change with temperature in currents and voltages in the voltage generator circuit **110**. FIG. 7 illustrates the power supply voltage **Vcc1**, the current **Ibg**, the current **Icc1** from the power supply line **L1a**, and the reference voltage **Vref1** when the temperature **T** [° C.] of the reference voltage circuit **53** is changed.

(76) In the voltage generator circuit **110**, the leakage current is generated from the bipolar devices provided in the reference voltage circuit **53**, in the temperature range above the temperature threshold value **Tth**. This increases the current **Ibg** flowing from the reference voltage circuit **53** toward the ground.

(77) The voltage generator circuit **110** includes no mechanism to limit an increase in the current **Icc1**. Accordingly, in response to the current **Ibg** being generated, the current **Icc1** supplied from the power supply line **L1a** to the reference voltage circuit **53** increases as well. At a predetermined temperature in the temperature range above the temperature threshold value **Tth**, the current value of the current **Icc1** exceeds the current value of the current **Im** in the voltage generator circuit **40a**.

(78) In the voltage generator circuit **110**, in accordance with an increase in the current **Ibg** with a rise in the temperature, the reference voltage **Vref1** from the reference voltage circuit **53** rises as well. Meanwhile, the voltage generator circuit **40a** in FIG. 2 can provide the reference voltage **Vref1** with less temperature dependence, as compared with the voltage generator circuit **110**.

Relationship Between **Im** In Embodiment Example and **Ibg** In Comparative Example

(79) FIG. 8 is an example of a schematic diagram illustrating change with temperature in the current **Im** of the voltage generator circuit **40a** and the current **Ibg** of the voltage generator circuit **110**. In FIG. 8, the translation of the current **Im** in the voltage generator circuit **40a** is given by a solid line, and the translation of the current **Ibg** of the voltage generator circuit **110** is given by a dashed-dotted line.

(80) In the voltage generator circuit **110**, in response to the leakage current being generated, the current **Ibg** also increases in the temperature range above the temperature threshold value **Tth**. Meanwhile, in the voltage generator circuit **40a**, the current **Ibg** is limited to the current **Im** in the range above the temperature threshold value **Tth** as well.

(81) The current **Im** in the voltage generator circuit **40a** according to an embodiment of the present disclosure is considered as current sufficient to operate the reference voltage circuit **53**. However, if the value of the current **Im** is increased excessively, the current **Ibg** cannot be limited when the current **Ibg** increases with the temperature becoming equal to or higher than the temperature threshold value **Tth**, for example.

(82) Thus, in an embodiment of the present disclosure, the current value of the current **Im** is determined such that the current value of the current **Ibg** in the voltage generator circuit **110** when the temperature is at the temperature threshold value **Tth** is the current value of the current **Im**, for example. With the value of the current **Im** being set as such, it is possible to restrain an increase in the leakage current in the reference voltage circuit **53** with reliability.

Modified Example of Embodiment Example

Configuration of Voltage Generator Circuit **40c**

(83) FIG. 9A illustrates an example of a circuit diagram of a voltage generator circuit **40c**. The following mainly describes differences between the voltage generator circuit **40c** and the voltage generator circuit **40a**. A bias current source **50b** of the voltage generator circuit **40c** includes a Zener diode **62**, a resistor **63**, and an MOS transistor **64**.

(84) The Zener diode **62** and the resistor **63** coupled in series are provided between the line **L1a** to which the power supply voltage **Vcc1** is applied and the ground. Here, the power supply voltage **Vcc1** is higher than the Zener voltage of the Zener diode **62**, and thus the gate voltage of the MOS transistor **64** results in the Zener voltage.

- (85) With provision of the Zener diode **62** as such, it is possible to apply steady-state voltage to the gate terminal of the MOS transistor **64** even when the power supply voltage V_{cc1} fluctuates.
- (86) The resistor **63** adjusts the current supplied from the power supply voltage V_{cc1} to the Zener diode **62**. The resistor **63** is electrically coupled between the gate terminal of the MOS transistor **64** and the power supply line $L1a$.
- (87) The MOS transistor **64** generates the bias current I_{bias} between the drain and source thereof, based on the voltage generated from the Zener diode **62**. Note that since the stable Zener voltage is applied from the Zener diode **62** to the MOS transistor **64**, the bias current I_{bias} outputted from the MOS transistor **64** is stabilized.
- (88) The Zener diode **62** according to an embodiment of the present disclosure is an example of a “second Zener diode”, and the MOS transistor **64** is an example of a “third MOS transistor”.

Configuration of Voltage Generator Circuit **40d**

- (89) FIG. **9B** illustrates an example of a circuit diagram of a voltage generator circuit **40d**. The following mainly describes differences between the voltage generator circuit **40d** and the voltage generator circuit **40a**. A bias current source **50c** of the voltage generator circuit **40d** further includes the diode-coupled depletion type MOS transistor **61**, the Zener diode **62**, the resistor **63**, and the MOS transistor **64**.
- (90) Similarly to the case of FIG. **9A**, the Zener diode **62** and the resistor **63** coupled in series are provided between the line $L1a$ to which the power supply voltage V_{cc1} is applied and the ground. Accordingly, the Zener voltage is applied to the MOS transistor **64**.
- (91) The depletion type MOS transistor **61** is diode-coupled, and thus functions as a device to cause the magnitude of the drain-source current of the MOS transistor **64** to be the desired bias current I_{bias} .

- (92) Further, in an embodiment of the present disclosure, the depletion type MOS transistor **61** and the MOS transistor **64** operate as so-called source follower. Thus, the voltage corresponding to the Zener voltage is applied to the drain of the depletion type MOS transistor **61**. Accordingly, even if the level of the power supply voltage V_{cc1} is high, the stable bias current I_{bias} can be generated.
- (93) The Zener diode **62** according to an embodiment of the present disclosure is another example of the “second Zener diode”, and the MOS transistor **64** is another example of the “third MOS transistor”.

Configuration of Voltage Generator Circuit **40e**

- (94) FIG. **10A** illustrates an example of a circuit diagram of a voltage generator circuit **40e**. The following mainly describes differences between the voltage generator circuit **40e** and the voltage generator circuit **40a**. The voltage generator circuit **40e** includes a diode **54** in place of the resistor **52**.
- (95) The diode **54** has an anode coupled to the current mirror circuit **51**, and a cathode coupled to the reference voltage circuit **53**. Assuming that a drop in the drain-source voltage in the MOS transistor **71** of the current mirror circuit **51** is V_{ds} , a difference $V_{cc1} - V_{ds}$ obtained by subtracting the voltage V_{ds} from the power supply voltage V_{cc1} is applied to the anode of the diode **54**.
- (96) From the cathode of the diode **54**, a voltage $V_{dd2} (=V_{cc1} - V_{ds} - 0.7 \text{ [V]})$, which is lower than $V_{cc1} - V_{ds}$ by a forward voltage (here, 0.7 V) is outputted. Even in such a case, the constant current I_m and the voltage V_{dd2} to operate the reference voltage circuit **53** are supplied to the reference voltage circuit **53**. Accordingly, the voltage generator circuit **40e** according to an embodiment of the present disclosure can also improve the temperature characteristics of the reference voltage circuit **53**.

Configuration of Voltage Generator Circuit **40f**

- (97) FIG. **10B** illustrates an example of a circuit diagram of a voltage generator circuit **40f**. The following mainly describes differences between the voltage generator circuit **40f** and the voltage generator circuit **40a**. The voltage generator circuit **40f** includes a Zener diode **55** without including the resistor **52**.

(98) In an embodiment of the present disclosure, the Zener diode **55** has a cathode coupled to the current mirror circuit **51**, and an anode that is grounded, such that the reference voltage circuit **53** and the Zener diode **55** are coupled in parallel. Accordingly, a Zener voltage V_{dd3} of the Zener diode **55** is supplied to the reference voltage circuit **53**.

(99) The voltage generator circuit **40f** as such can lower the voltage to operate the reference voltage circuit **53** while supplying constant current to the reference voltage circuit **53**. Accordingly, in an embodiment of the present disclosure, even when the temperature changes, the reference voltage V_{ref1} from the reference voltage circuit **53** can be stabilized.

(100) Even in this embodiment of the present disclosure as well, the voltage generator circuit **40f** results in a circuit in which the current I_m being constant current and the voltage V_{dd3} being constant voltage are inputted to the reference voltage circuit **53**, and the temperature characteristics of the reference voltage circuit **53** are stabilized. The Zener diode **55** according to an embodiment of the present disclosure corresponds to a “first Zener diode”.

Configuration of Voltage Generator Circuit **40g**

(101) FIG. **10C** illustrates an example of a circuit diagram of a voltage generator circuit **40g**. The following mainly describes differences between a voltage generator circuit **40g** and the voltage generator circuit **40a**.

(102) In the voltage generator circuit **40g**, the reference voltage circuit **53** is directly electrically coupled to the output from the MOS transistor **71** out of the outputs of the current mirror circuit **51**.

(103) Even without the resistor **52** being provided, the current I_m from the current mirror circuit **51** to the reference voltage circuit **53** is limited, and thus change with temperature in the reference voltage V_{ref1} of the voltage generator circuit **40g** is smaller than change with temperature in the reference voltage V_{ref1} of the voltage generator circuit **110**. In other words, in the voltage generator circuit **40g** according to an embodiment of the present disclosure, the temperature characteristics are improved as compared with the voltage generator circuit **110** in FIG. **6**.

(104) Further, the current mirror circuit **51** has an internal resistance, and thus by appropriately setting the internal resistance of the current mirror circuit **51**, it is possible to lower voltage V_m outputted from the current mirror circuit **51**. Accordingly, when the internal resistance is adjustable, the same effects can be achieved according to this embodiment of the present disclosure as well, as in such an embodiment including the resistor **52**.

SUMMARY

(105) Hereinabove, the semiconductor module **10** according to an embodiment of the present disclosure has been explained. For example, as illustrated in FIG. **1**, the integrated circuit **22a** of the semiconductor module **10** includes the voltage generator circuit **40a** electrically coupled to the power supply line $L1a$ to which the power supply voltage V_{cc1} is to be applied.

(106) For example, as illustrated in FIG. **2**, in the voltage generator circuit **40a**, the current mirror circuit **51** supplies constant current to the reference voltage circuit **53**. Accordingly, in the voltage generator circuit **40a** of the integrated circuit **22a**, the voltage value of the reference voltage V_{ref1} outputted from the reference voltage circuit **53** is stabilized even when the temperature is high with a small circuit configuration, thereby being able to improve the temperature characteristics of the reference voltage circuit **53**.

(107) For example, as illustrated in FIG. **2**, the voltage generator circuit **40a** may include the resistor **52** to supply the voltage V_{dd1} to the reference voltage circuit **53**. Accordingly, in the voltage generator circuit **40a** of the integrated circuit **22a**, the voltage V_{dd1} to be inputted to the reference voltage circuit **53** results in a low voltage, thereby being able to further improve the temperature characteristics of the reference voltage circuit **53**.

(108) For example, as illustrated in FIG. **10A**, the voltage generator circuit **40e** may include the diode **54** to supply the voltage V_{dd2} to the reference voltage circuit **53**. Accordingly, in the voltage generator circuit **40e** of the integrated circuit **22a**, the voltage exceeding the forward voltage is applied to the diode **54**, thereby being able to improve the temperature characteristics of the

reference voltage circuit **53**.

(109) For example, as illustrated in FIG. **10B**, the voltage generator circuit **40f** may include the Zener diode **55** to supply the voltage V_{dd3} to the reference voltage circuit **53**. Accordingly, in the voltage generator circuit **40f** of the integrated circuit **22a**, the temperature characteristics of the reference voltage circuit **53** can be improved in a range in which the Zener diode **55** can be maintained in a breakdown state.

(110) For example, as illustrated in FIG. **3**, the reference voltage circuit **53** may be a bandgap circuit including bipolar devices such as the bipolar transistor **86** and the bipolar transistor **87**. In the reference voltage circuit **53** as such, the surface area of the PN junction portion is large as illustrated in FIG. **4**, and thus it is effective to reduce the leakage current.

(111) For example, as illustrated in FIG. **5**, the current I_m is determined such that the current I_{bg} is limited to the value of the current I_m . This can stabilize the reference voltage V_{ref1} , thereby being able to improve the temperature characteristics of the reference voltage circuit **53**.

(112) For example, as illustrated in FIG. **7**, when the temperature is higher than the temperature threshold value T_{th} , the current value of the current I_m supplied by the current mirror circuit **51** of the voltage generator circuit **40a** to the reference voltage circuit **53** is smaller than the current value of the current I_{cc1} supplied from the power supply line **L1a** to the reference voltage circuit **53** in a state where the reference voltage circuit **53** is coupled to the power supply line **L1a** as in the voltage generator circuit **110**.

(113) As such, the voltage generator circuit **40a** of the integrated circuit **22a** can be designed such that the current supplied to the reference voltage circuit **53** is small.

(114) For example, as illustrated in FIG. **2**, the voltage generator circuit **40a** of the integrated circuit **22a** may include the bias current source **50a**. In this case, the constant current source of the voltage generator circuit **40a** may be the current mirror circuit **51** including the MOS transistor **70** through which the bias current I_{bias} of the bias current source **50a** is to flow, and the MOS transistor **71** to supply the constant current I_m to the reference voltage circuit **53**, without including a bipolar transistor.

(115) Accordingly, in the voltage generator circuit **40a** of the integrated circuit **22a**, the leakage current is less likely to be generated from the current mirror circuit **51**, which makes it easier to supply constant current to the reference voltage circuit **53**. Accordingly, the temperature characteristics of the reference voltage circuit **53** can be improved.

(116) For example, as illustrated in FIG. **9A** or **9B**, the bias current source **50** may include the Zener diode **62** and the MOS transistor **64** to generate the bias current I_{bias} , based on the voltage generated at the Zener diode **62**.

(117) Accordingly, the voltage generator circuit **40a** of the integrated circuit **22a** can provide the stable bias current I_{bias} even when the power supply voltage V_{cc1} fluctuates.

(118) For example, as illustrated in FIG. **2** or **9B**, the bias current source **50** may include the diode-coupled depletion type MOS transistor **61**. This makes it possible for the voltage generator circuit **40a** or **40d** of the integrated circuit **22a** to generate the desired bias current I_{bias} even at a low voltage with a small circuit configuration.

(119) For example, as illustrated in FIG. **3**, the reference voltage circuit **53** may be such a circuit that includes the current source **88**, the current source **89**, the bipolar transistor **86** to which the current from the current source **88** is to be supplied, the resistor **85** to which the current from the current source **89** is to be supplied, the bipolar transistor **87** to be electrically coupled to the resistor **85**, and that is configured to output the temperature-compensated reference voltage V_{ref1} .

(120) In this case, the voltage generator circuit **40a** of the integrated circuit **22a** can output such reference voltage V_{ref1} that the effects of respective temperature coefficients of the bipolar transistors **86** and **87** are compensated in a temperature range in which no leakage current is generated from the bipolar transistors **86** and **87**.

(121) For example, as illustrated in FIG. **1**, the semiconductor module **10** includes an integrated

circuit **22** according to an embodiment of the present disclosure. This makes it possible to utilize the reference voltage **Vref1** of the reference voltage circuit **53** with temperature characteristics thereof improved, in the semiconductor module **10**.

(122) The present disclosure is directed to provision of an integrated circuit capable of stabilizing an output of a reference voltage circuit even when a temperature rises.

(123) It is possible to provide an integrated circuit capable of stabilizing an output of a reference voltage circuit even when a temperature rises.

(124) Hereinabove, the present disclosure has been described using embodiments. However, the technical scope of the present disclosure is not limited to the range described in above embodiments. It is apparent to those skilled in the art that above embodiments can be variously altered and modified. It is apparent from the claims that the technical scope of the present disclosure includes such altered or modified modes and equivalents thereof without departing from its essential features of the present disclosure.

(125) It should be noted that processes such as operations, procedures, steps, stages, and the like in a device, a system, a program, and a method described in the claims, the specification, and the drawings may be performed in any order, unless a term such as “before”, “prior to” or the like is explicitly used or an output of a previous process is used in a subsequent process. Even if terms such as “first”, “next”, and/or the like is used, for convenience, with respect to an operation flow in the claims, the specification, and the drawings, this does not mean that the flow needs to be performed in this order.

Claims

1. An integrated circuit comprising: a power supply line configured to receive a power supply voltage; a constant current source electrically coupled to the power supply line; a reference voltage circuit electrically coupled to the constant current source; and a first resistor having two ends, one end thereof being electrically coupled to the constant current source, and the other end thereof being electrically coupled to the reference voltage circuit, wherein the reference voltage circuit is a bandgap circuit including a plurality of bipolar devices, and the first resistor is configured to decrease a leakage current in the plurality of bipolar devices when a temperature rises.
2. The integrated circuit according to claim 1, wherein a current value of the constant current source is determined such that a current value of a current flowing from the reference voltage circuit to a ground is limited to a current value of the constant current source, when the temperature is equal to or higher than a predetermined temperature at which the leakage current is generated.
3. The integrated circuit according to claim 1, wherein a current value of a first current, which is supplied to the reference voltage circuit by the constant current source when the temperature is a predetermined temperature, is smaller than a current value of a second current, which is supplied from the power supply line to the reference voltage circuit when the temperature is the predetermined temperature, in a state where the reference voltage circuit is coupled to the power supply line.
4. The integrated circuit according to claim 1, wherein the integrated circuit includes a bias current source, and the constant current source includes a first metal-oxide-semiconductor (MOS) transistor allowing a bias current of the bias current source to flow therethrough, and a second MOS transistor configuring a current mirror circuit with the first MOS transistor, the second MOS transistor being configured to supply a constant current to the reference voltage circuit.
5. The integrated circuit according to claim 4, wherein the bias current source includes a second Zener diode, and a third MOS transistor configured to generate the bias current, based on a voltage generated at the second Zener diode.
6. The integrated circuit according to claim 4, wherein the bias current source includes a diode-coupled depletion type MOS transistor.

7. The integrated circuit according to claim 1, wherein the reference voltage circuit includes a first current source configured to receive a current from the constant current source, a second current source configured to receive the current from the constant current source, the second current source being electrically coupled in parallel with the first current source, a first bipolar transistor electrically coupled in series to the first current source, a second resistor electrically coupled in series to the second current source, and a second bipolar transistor electrically coupled in series to the second resistor, and the reference voltage circuit is configured to output a voltage at the second resistor as a temperature-compensated reference voltage.

8. A semiconductor module, comprising the integrated circuit according to claim 1.

9. An integrated circuit comprising: a power supply line configured to receive a power supply voltage; a constant current source electrically coupled to the power supply line; a reference voltage circuit electrically coupled to the constant current source; and a diode having an anode electrically coupled to the constant current source, and a cathode electrically coupled to the reference voltage circuit, wherein the integrated circuit includes a bias current source, and the constant current source includes a single first metal-oxide-semiconductor (MOS) transistor allowing a bias current of the bias current source to flow therethrough, and a single second MOS transistor configuring a current mirror circuit with the single first MOS transistor, the single second MOS transistor being configured to supply a constant current to the anode of the diode; and the reference voltage circuit includes a first current source configured to receive a current from the constant current source, a second current source configured to receive the current from the constant current source, the second current source being electrically coupled in parallel with the first current source, a first bipolar transistor electrically coupled in series to the first current source, a second resistor electrically coupled in series to the second current source, and a second bipolar transistor electrically coupled in series to the second resistor, the reference voltage circuit being configured to output a voltage at the second resistor as a temperature-compensated reference voltage.

10. An integrated circuit comprising: a power supply line configured to receive a power supply voltage; a constant current source electrically coupled to the power supply line; and a reference voltage circuit and a first Zener diode connected in parallel, both being electrically coupled between the constant current source and a ground, wherein the first Zener diode has a cathode electrically coupled to the reference voltage circuit and the constant current source, and an anode that is grounded, such that a Zener voltage of the first Zener diode is supplied directly to the reference voltage circuit as a voltage to operate the reference voltage circuit, and the reference voltage circuit is a bandgap circuit including a plurality of bipolar devices, the reference voltage circuit generating a reference voltage based on the Zener voltage.

11. The integrated circuit of claim 10, further comprising: a bias current source, wherein the constant current source includes a first metal-oxide-semiconductor (MOS) transistor allowing a bias current of the bias current source to flow therethrough, and a second MOS transistor configuring a current mirror circuit with the first MOS transistor, the second MOS transistor being configured to supply a constant current to the reference voltage circuit.
